

Chikungunya transmission and population susceptibility in Bahia, Brazil (2015-2025)

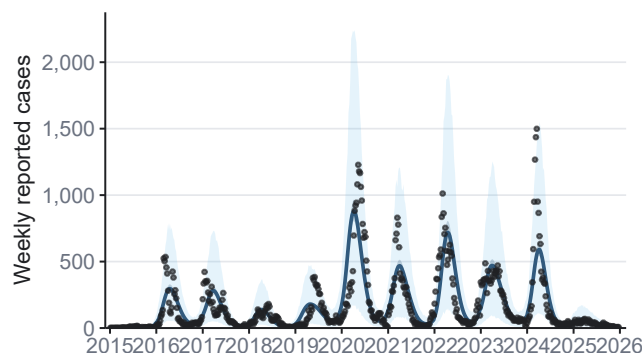
Frozen Ceara v4.9 structure, external replication, $q = 0.05$ (PILOT, HMC PASS) | case likelihood only

A

Reported cases and posterior prediction

No seed mechanism used (no Bahia gap qualifies -- s_i)

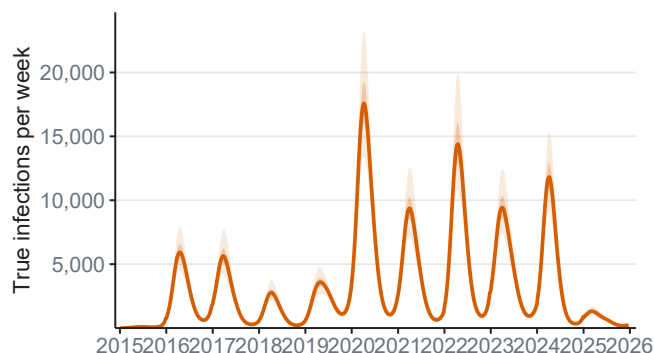
— Model expectation • Observed cases



B

Latent infection incidence

Median, 50% and 95% credible intervals

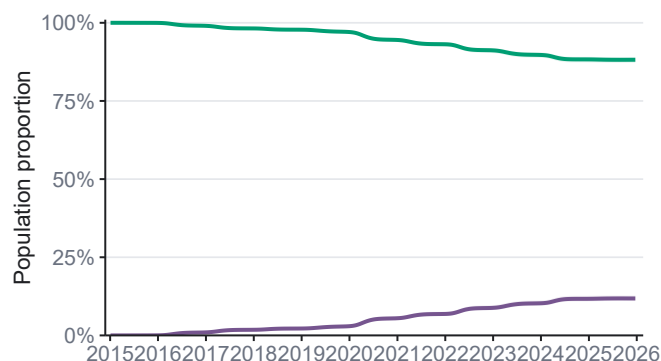


C

Population susceptibility and immunity

Closed population; LIFELONG immunity; conditional on

— Infection-derived immune — Susceptible

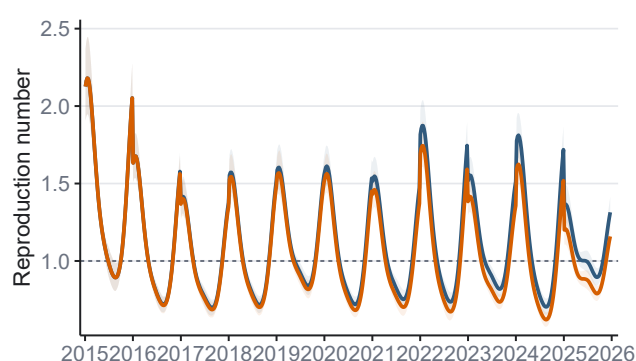


D

Time-varying reproduction numbers

Frozen v4.9 structure: harmonic + year-specific amplitude

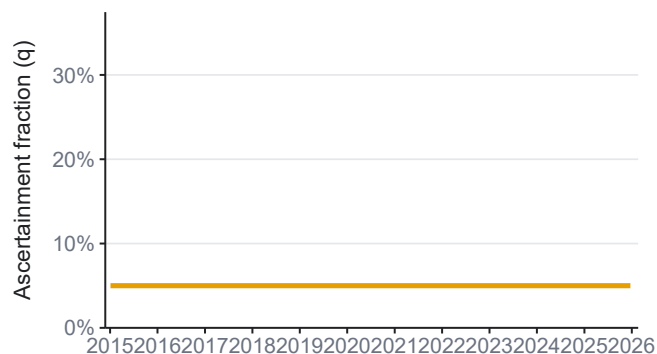
— $R_0(t)$ — $R_{eff}(t)$



E

Case ascertainment

q FIXED at 0.05 (not estimated; no Bahia serology -- cc)



F

Cumulative infection (immune fraction)

NO serology anchor for Bahia -- not borrowed from Ceara

